



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 1 REPORT

Internship Program — Batch 2026

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Submitted To:	Trainer Name/
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1. Introduction

This report presents the work completed for **Task 1** of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. The internship focuses on practical exposure to core concepts in Artificial Intelligence and Machine Learning, equipping interns with skills applicable to real-world technology and marketing challenges.

M-Tech Production & Marketing is a forward-looking company based in Haripur, KPK, specializing in Technology, Training, Marketing, and Projects. The internship is structured in a 9-Phase roadmap designed to progressively build theoretical knowledge and hands-on project experience.

2. Objectives

- To design and develop a desktop-based Hostel Room Allocation System that assigns hostel rooms to students in a fair, efficient, and constraint-aware manner.
- To apply Artificial Intelligence problem-solving techniques, specifically a Constraint Satisfaction Problem (CSP) approach, to a real-world resource allocation scenario.
- To incorporate a Preference Matching Algorithm that considers roommate compatibility (gender and lifestyle preferences) alongside mandatory constraints such as room capacity and accessibility requirements.
- To build a complete, functional system covering student management, room management, automated allocation, and result visualization using Python, Tkinter, and SQLite.

3. Task Description

The assigned task was to design and implement a Hostel Room Allocation System capable of automatically assigning hostel rooms to students while satisfying key institutional and human-centric constraints. The system needed to:

- Maintain student records, including ID, name, gender, department, semester, accessibility requirement, and roommate preference.
- Maintain room records, including room number, capacity, current occupancy, and accessibility support.
- Automatically allocate unallocated students to available rooms using an algorithm that respects hard constraints (room capacity and accessibility needs) while optimizing soft constraints (gender and lifestyle preference compatibility).
- Display allocation results with a calculated compatibility score for every student-room assignment.
- Provide a clean, professional desktop interface for all of the above operations, built entirely with Python's standard library.

4. Work Done

The project was developed step by step, following a modular folder structure that separates the database layer, data models, allocation logic, and user interface:

- Designed an SQLite database with three tables - Students, Rooms, and Allocations - along with a data-access layer (db_manager.py) providing full CRUD operations.
- Created Student and Room data model classes to keep the allocation logic clean and readable.
- Implemented the core allocation engine (algorithms/allocation.py) combining a Constraint Satisfaction Problem approach (hard constraints: room capacity and accessibility) with a Preference Matching Algorithm (soft constraints: gender and lifestyle preference), producing a 0-100 compatibility score for each allocation.
- Built a desktop GUI using Tkinter with five main screens: Dashboard, Student Management, Room Management, Room Allocation, and View Allocations.
- Styled the interface with a consistent dark color theme and added simple animations - a sliding title, hover-animated buttons, an animated progress bar during allocation, and a pulsing success banner - to make the application feel polished and responsive.
- Tested the complete workflow end-to-end: adding students and rooms, running the allocation algorithm, verifying that accessibility-requiring students were correctly placed into accessibility-supported rooms, and viewing/resetting allocation results.

5. Challenges Faced

- The GUI was initially built using PyQt5, but installation and runtime compatibility issues on the development machine required migrating the entire front end to Tkinter, without altering the underlying database or algorithm logic.
- Designing a scoring formula that balances multiple soft constraints (gender match and preference match) in a way that is simple, transparent, and easy to explain and justify.
- Ensuring that students with accessibility requirements are never blocked out of the only suitable room, which required processing them before regular students during the allocation step.
- Keeping the interface visually engaging (colors and animations) while using only Python's standard library, to avoid introducing unnecessary external dependencies.

6. Key Learnings

- Gained hands-on experience applying Constraint Satisfaction Problem (CSP) concepts to a real, human-centric allocation problem rather than a purely theoretical exercise.
- Learned how to design a lightweight, explainable preference-matching scoring system for roommate compatibility.
- Understood the importance of separating a project into clear layers - database, models, algorithms, and interface - for maintainability and clarity.
- Learned to build, style, and animate a desktop GUI using Tkinter, including hover effects, progress indicators, and color transitions.
- Improved practical skills in SQLite database design, debugging, and iteratively refactoring a working application in response to changing requirements.

7. Conclusion

The Hostel Room Allocation System successfully demonstrates how a classical AI technique - Constraint Satisfaction - can be combined with a preference-matching algorithm to solve a practical, everyday institutional problem. The completed system fulfills all functional requirements: student and room

management, automated fair allocation, and clear result visualization, built with a clean, modular, and maintainable codebase. This task strengthened my understanding of applying AI problem-solving approaches to real-world constraints and improved my practical software development skills using Python, Tkinter, and SQLite.

Intern Signature: _____

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Date: June 2026

Supervisor: _____

Name:

Designation: